

Lectures on Machine Learning

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2026-01-01

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Preface

These notes are being developed for the UConn Math Department's undergraduate course on the Mathematics of Machine Learning, Math 3180.

$$\int_0^\infty f(x)dx$$

This is a (very) *rough draft*.

You can view this project on GitHub. Please let the author know if you have questions or find mistakes by opening an issue there.

1 Linear Regression

1.1 Introduction

Suppose that we are trying to study two quantities x and y that we suspect are related – at least approximately – by a linear equation $y = ax + b$. Sometimes this linear relationship is predicted by theoretical considerations, and sometimes it is just an empirical hypothesis.

For example, if we are trying to determine the velocity of an object travelling towards us at constant speed, and we measure the distances $d_1, d_2, \dots, d_n$ between us and the object at a series of times $t_1, t_2, \dots, t_n$, then since “distance equals rate times time” we have a theoretical foundation for the assumption that $d = rt + b$ for some constants r and b . On the other hand, because of unavoidable experimental errors, we can’t expect that this relationship will hold exactly for the observed data; instead, we likely get a graph like that shown in Figure ???. We’ve drawn a line on the plot that seems to capture the true slope (and hence velocity) of the object.

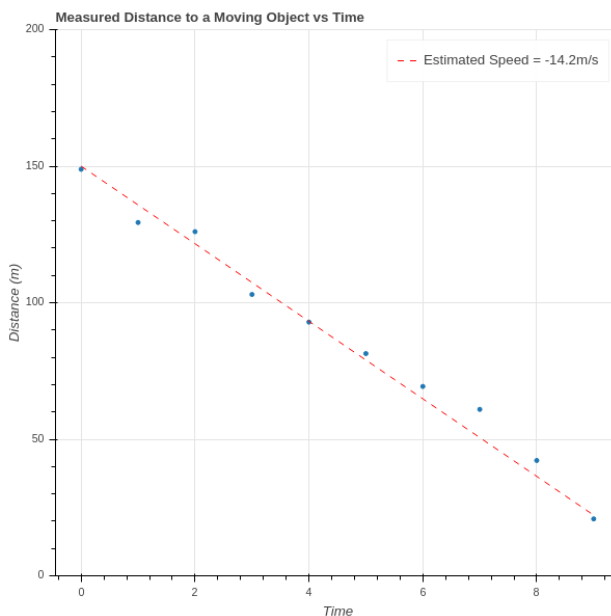


Figure 1.1: Physics Experiment

On the other hand, we might look at a graph such as Figure ??, which plots the gas mileage of various car models against their engine size (displacement), and observe a general trend in which bigger engines get lower mileage. In this situation we could ask for the best line of the form $y = mx + b$ that captures this relationship and use that to make general conclusions without necessarily having an underlying theory.

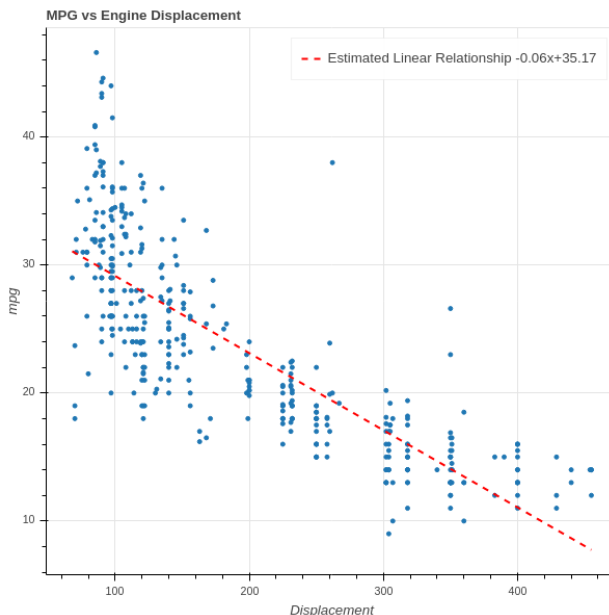


Figure 1.2: MPG vs Displacement ([1])

1.2 Least Squares (via Calculus)

In either of the two cases above, the question we face is to determine the line $y = mx + b$ that “best fits” the data $\{(x_i, y_i)_{i=1}^N\}$. The classic approach is to determine the equation of a line $y = mx + b$ that minimizes the “mean squared error”:

$$MSE(m, b) = \frac{1}{N} \sum_{i=1}^N (y_i - mx_i - b)^2$$

It’s worth emphasizing that the MSE is a function of two variables – the slope m and the intercept b – and that the data points $\{(x_i, y_i)\}$ are constants for these purposes. Furthermore, it’s a quadratic function in those two variables. Since our goal is to find m and b that minimize the MSE , we have a Calculus problem that we can solve by taking partial derivatives and setting them to zero.

To simplify the notation, let’s abbreviate MSE by E .

$$\begin{aligned} \frac{\partial E}{\partial m} &= \frac{1}{N} \sum_{i=1}^N -2x_i(y_i - mx_i - b) \\ \frac{\partial E}{\partial b} &= \frac{1}{N} \sum_{i=1}^N -2(y_i - mx_i - b) \end{aligned}$$

We set these two partial derivatives to zero, so we can drop the -2 and regroup the sums to obtain two equations in two unknowns (we keep the $\frac{1}{N}$ because it is illuminating in the final result):

$$\begin{aligned}
\frac{1}{N} \left(\sum_{i=1}^N x_i^2 \right) m + \frac{1}{N} \left(\sum_{i=1}^N x_i \right) b &= \frac{1}{N} \sum_{i=1}^N x_i y_i \\
\frac{1}{N} \left(\sum_{i=1}^N x_i \right) m + b &= \frac{1}{N} \sum_{i=1}^N y_i
\end{aligned} \tag{1.1}$$

In these equations, notice that $\frac{1}{N} \sum_{i=1}^N x_i$ is the average (or mean) value of the x_i . Let's call this $\bar{x}$. Similarly, $\frac{1}{N} \sum_{i=1}^N y_i$ is the mean of the y_i , and we'll call it $\bar{y}$. If we further simplify the notation and write S_{xx} for $\frac{1}{N} \sum_{i=1}^N x_i^2$ and S_{xy} for $\frac{1}{N} \sum_{i=1}^N x_i y_i$ then we can write down a solution to this system using Cramer's rule:

$$\begin{aligned}
m &= \frac{S_{xy} - \bar{x}\bar{y}}{S_{xx} - \bar{x}^2} \\
b &= \frac{S_{xx}\bar{y} - S_{xy}\bar{x}}{S_{xx} - \bar{x}^2}
\end{aligned} \tag{1.2}$$

where we must have $S_{xx} - \bar{x}^2 \neq 0$.

1.2.1 Exercises

1. Verify that Equation ?? is in fact the solution to the system in Equation ?? .
2. Suppose that $S_{xx} - \bar{x}^2 = 0$. What does that mean about the x_i ? Does it make sense that the problem of finding the “line of best fit” fails in this case?

1.3 Least Squares (via Geometry)

In our discussion above, we thought about our data as consisting of N pairs (x_i, y_i) corresponding to N points in the xy -plane $\mathbf{R}^2$. Now let's turn that picture “on its side”, and instead think of our data as consisting of *two* points in $\mathbf{R}^N$:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ and } Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Let's also introduce one other vector

$$E = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

1 Linear Regression

First, let's assume that E and X are linearly independent. If not, then X is a constant vector (why?) which we already know is a problem from Section ??, Exercise 2. Therefore E and X span a plane in $\mathbf{R}^N$.

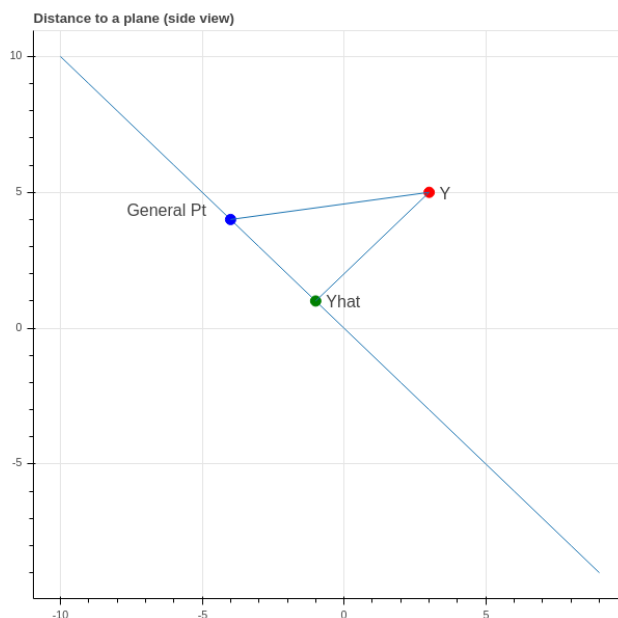


Figure 1.3: Distance to A Plane

Now if our data points (x_i, y_i) all *did* lie on a line $y = mx + b$, then the three vectors X , Y , and E would be linearly dependent:

$$Y = mX + bE.$$

Since our data is only approximately linear, that's not the case. So instead we look for an approximate solution. One way to phrase that is to ask:

What is the point $\hat{Y}$ in the plane H spanned by X and E in $\mathbf{R}^N$ which is closest to Y ?

If we knew this point $\hat{Y}$, then since it lies in H we would have $\hat{Y} = mX + bE$ and the coefficients m and b would be a candidate for defining a line of best fit $y = mx + b$. Finding the point in a plane closest to another point in $\mathbf{R}^N$ is a geometry problem that we can solve.

Proposition: The point $\hat{Y}$ in the plane spanned by X and E is the point such that the vector $Y - \hat{Y}$ is perpendicular to H .

Proof: See Figure ?? for an illustration – perhaps you are already convinced by this, but let's be careful. $\hat{Y} = mX + bE$ such that

$$D = \|Y - \hat{Y}\|^2 = \|Y - mX - bE\|^2$$

is minimal. Using some vector calculus, we have

$$\frac{\partial D}{\partial m} = \frac{\partial}{\partial m} (Y - mX - bE) \cdot (Y - mX - bE) = -2(Y - mX - bE) \cdot X$$

and

$$\frac{\partial D}{\partial b} = \frac{\partial}{\partial b}(Y - mX - bE) \cdot (Y - mX - bE) = -2(Y - mX - bE) \cdot E.$$

So both derivatives are zero exactly when $\hat{Y} = (Y - mX - bE)$ is orthogonal to both X and E , and therefore every vector in H .

We also obtain equations for m and b just as in our first look at this problem.

$$\begin{aligned} m(X \cdot E) + b(E \cdot E) &= (Y \cdot E) \\ m(X \cdot X) + b(E \cdot X) &= (Y \cdot X) \end{aligned} \tag{1.3}$$

We leave it is an exercise below to check that these are the same equations that we obtained in Equation ??.

1.3.1 Exercises

1. Verify that Equation ?? and Equation ?? are equivalent.

1.4 The Multivariate Case (Calculus)

Having worked through the problem of finding a “line of best fit” from two points of view, let’s look at a more general problem. We looked above at a scatterplot showing the relationship between gas mileage and engine size (displacement). There are other factors that might contribute to gas mileage that we want to consider as well – for example:

- a car that is heavy compared to its engine size may get worse mileage
- a sports car with a drive train that gives fast acceleration as compared to a car with a transmission designed for long trips may have different mileage for the same engine size.

Suppose we wish to use engine displacement, vehicle weight, and acceleration all together to predict mileage. Instead of looking points (x_i, y_i) where x_i is the displacement of the i^{th} car model and we try to predict a value y from a corresponding x as $y = mx + b$ – let’s look at a situation in which our measured value y depends on multiple variables – say displacement d , weight w , and acceleration a with $k = 3$ – and we are trying to find the best linear equation

\$\$

$$y = m_1 d + m_2 w + m_3 a + b$$

\$\$ \{\text{\#eq-multivariate}\}

But to handle this situation more generally we need to adopt a convention that will allow us to use indexed variables instead of d , w , and a . We will use the *tidy* data convention.

Tidy Data: A dataset is tidy if it consists of values x_{ij} for $i = 1, \dots, N$ and $j = 1, \dots, k$ so that:

- the row index corresponds to a *sample* – a set of measurements from a single event or item;

1 Linear Regression

- the column index corresponds to a *feature* – a particular property measured for all of the events or items.

In our case,

- the *samples* are the different types of car models,
- the *features* are the properties of those car models.

For us, N is the number of different types of cars, and k is the number of properties we are considering. Since we are looking at displacement, weight, and acceleration, we have $k = 3$.

So the “independent variables” for a set of data that consists of N samples, and k measurements for each sample, can be represented by a $N \times k$ matrix

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1k} \\ x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{Nk} \end{pmatrix}$$

and the measured dependent variables Y are a column vector

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}.$$

If $m_1, \dots, m_k$ are “slopes” associated with these properties in **Eq-multivariate**, and b is the “intercept”, then the predicted value $\hat{Y}$ is given by a matrix equation

$$\hat{Y} = X \begin{bmatrix} m_1 \\ m_2 \\ \dots \\ m_k \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ \dots \\ 1 \end{bmatrix} b$$

and our goal is to choose these parameters m_i and b to make the mean squared error:

$$MSE(m_1, \dots, m_k, b) = \|Y - \hat{Y}\|^2 = \sum_{i=1}^N (y_i - \sum_{j=1}^k x_{ij} m_j - b)^2.$$

Here we are summing over the N different car models, and for each model taking the squared difference between the true mileage y_i and the “predicted” mileage $\sum_{j=1}^k x_{ij} m_j + b$. We wish to minimize this MSE.

Let’s make one more simplification. The intercept variable b is annoying because it requires separate treatment from the m_i . But we can use a trick to eliminate the need for special treatment. Let’s add a new feature to our data matrix (a new column) that has the constant value 1.

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1k} & 1 \\ x_{21} & x_{22} & \cdots & x_{2k} & 1 \\ \vdots & \vdots & \ddots & \vdots & 1 \\ x_{N1} & x_{N2} & \cdots & x_{Nk} & 1 \end{pmatrix}$$

Now our data matrix X is $N \times (k+1)$ and we can put our “intercept” $b = m_{k+1}$ into our vector of “slopes” $m_1, \dots, m_k, m_{k+1}$:

$$\hat{Y} = X \begin{bmatrix} m_1 \\ m_2 \\ \vdots \\ m_k \\ m_{k+1} \end{bmatrix}$$

and our MSE becomes

$$MSE(M) = \|Y - XM\|^2$$

where

$$M = \begin{bmatrix} m_1 \\ m_2 \\ \vdots \\ m_k \\ m_{k+1} \end{bmatrix}.$$

Remark: Later on (see {Section ??}) we will see that if we “center” our features about their mean, by subtracting the average value of each column of X from that column; and we also subtract the average value of Y from the entries of Y , then the b that emerges from the least squares fit is zero. As a result, instead of adding a column of 1’s, you can change coordinates to center each feature about its mean, and keep your X matrix $N \times k$.

The Calculus approach to minimizing the MSE is to take its partial derivatives with respect to the m_i and set them to zero. Let’s first work out the derivatives in a nice form for later.

Proposition: The gradient of $MSE(M) = E$ is given by

$$\nabla E = \begin{bmatrix} \frac{\partial}{\partial m_1} E \\ \frac{\partial}{\partial m_2} E \\ \vdots \\ \frac{\partial}{\partial m_{M+1}} E \end{bmatrix} = -2X^\top Y + 2X^\top XM \quad (1.4)$$

where $X^\top$ is the transpose of X .

Proof: